



Advanced Corporate Finance

BUSINESS VALUATION OF MERCK & CO.

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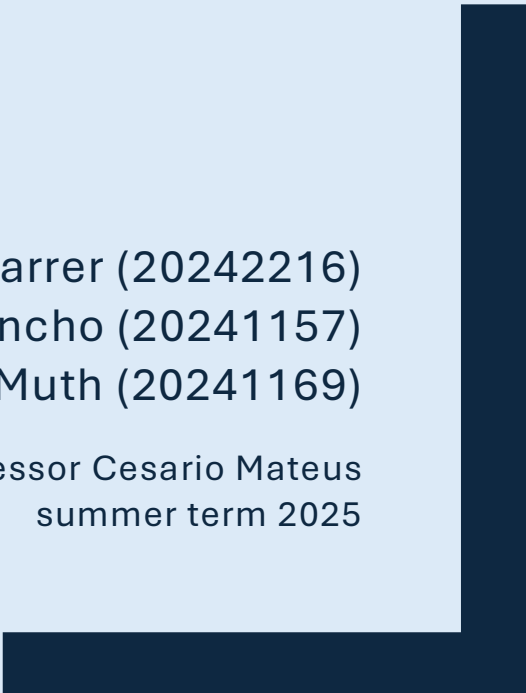


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Recommendation BUY

Merck & Co., Inc.

Target Price \$122.61

Current Price* \$78.39

Upside 56.41 %

*cutoff date: 15.04.2025

Industry Pharmacy

Sector Health Care

Ticker Symbol MRK

Stock Exchange NYSE

1. Investment Summary

Merck & Co presents a strong investment opportunity underpinned by its robust market position, high-margin business model, and forward-looking strategic direction. The company is currently rated a Buy, with a target price of \$122.61, representing an estimated 56% upside from its closing market price on April 15, 2025. This valuation is supported by both a Free Cash Flow to Equity (FCFE) model and a Dividend Discount Model (DDM), which yield's fair value estimates of \$126.93 and \$105.22 respectively. Even under conservative (bear case) assumptions, the stock is valued at \$113.74, while the bull case scenario reaches \$144.56, demonstrating that the company appears fundamentally undervalued across a range of outcomes.

At the heart of Merck's value proposition is its flagship oncology drug, Keytruda, which accounts for approximately 50% of total revenues. With a CAGR of just over 20% in recent years, Keytruda continues to drive growth and is expected to remain a key revenue generator through to 2028. Whilst the drug's patent is set to expire that year, Merck is already developing a subcutaneous version with launch targeted for 2026, aimed at extending exclusivity and market share. Beyond oncology, Merck benefits from a diversified portfolio that includes a strong presence in vaccines and a unique competitive edge in animal health, an area largely untapped by other major pharmaceutical players.

Financially, Merck & Co is in excellent health. It boasts a gross margin of 77.1%, which is expected to rise to 82.5% in 2025. This metric outperforms both peers and the industry average of 69.3%. Its Return on Equity (36.9%) and Return on Assets (14.6%) signal efficient use of capital and asset management, while its P/E ratio of 11.95

suggests the stock is trading at a discount relative to an industry average of 26.8. Liquidity is also strong, with a current ratio of 1.36. Together, these metrics reflect a financially sound company well-positioned for long-term growth.

Merck is also navigating macroeconomic and regulatory risks with clear foresight. Although Gardasil, its second-largest revenue driver, saw declining sales in China in quarter 4, 2024, management remains optimistic about long-term demand due to low vaccination rates. The company is also managing potential tariff exposure by investing \$8 billion into domestic manufacturing facilities, effectively hedging against global trade disruptions. Con-currently, the firm is also subject to a class action lawsuit, failing to disclose falling Gardasil sales in China.

On the Environmental, Social, and Governance (ESG) front, Merck scores a 52/100 on the S&P Global scale, above the industry average of 35, and demonstrates strong governance through independent board committees, performance-based executive compensation, and active shareholder engagement. While Merck has made progress toward its climate goals, including reducing Scope 1 and 2 emissions, Scope 3 emissions have increased, suggesting room for improvement in its broader sustainability strategy. Commenting on governance, the board of directors boasts significant industry experience but also tenure at Merck. However, their latest actions failing to update the market to price sensitive information is questionable.

In summary, Merck & Co combines strong fundamentals, sector leadership, high-margin operations, and a forward-thinking growth strategy. Its attractive valuation relative to peers, high gross margins and its resilient revenue base make it a compelling investment. For investors seeking exposure to the healthcare sector, with a stalwart company, Merck & Co offers good value.

2. Business Description

2.1 Company history

Merck & Co is a leading biopharmaceutical company. The company is research-intensive, aiming to be a leading company that offers solutions to the treatment and prevention of diseases in both humans and animals. The business model is based on generating revenue from the sale of pharmaceutical products to treat disease within humans and animals. Segregating, sales stem from three dominant areas, Oncology,

Vaccines and Animal health. Merck & Co's mission is to "use the power of leading-edge science to save and improve lives around the world".

Merck was founded in 1891 and first delivered chemicals throughout New York during this time. From the beginning of the 1900's Merck & Co began to investigate treatments and increase their footsteps within the pharmaceutical research industry. Subsequently Merck became the first synthesizers and producers of the 'B vitamin' and committed to making vitamins more readily available.

The 1940's and onwards saw Merck make breakthroughs in antibiotics that fought Tuberculosis, chronic diseases, and parasitic poultry disease. After merging with Sharpe & Dohme who were a chemical research and manufacturing company, Merck & Co continued to innovate, finding solutions to treat high blood pressure and heart disease and measles-mumps vaccines. This was later followed by the development of a vaccine for Pneumonia.

More recently Merck & Co has been involved in the treatment of HIV in 1996 and the prevention and treatment of asthma in 1998. In 2009 Merck & Co and Schering-Plough merged to become the second largest pharmaceutical company in the United States. More recently Merck & Co continue to research and innovate, by coming up with a vaccine for Ebola virus but also developed chewable dog tablets that kill fleas and ticks within the animal health division.

2.2 Today's business

Merck & Co today operates as a globally integrated biopharmaceutical company, driven by science and committed to delivering innovative health solutions. The business is structured around three principal segments: Human Health, Animal Health, and Manufacturing & Research Infrastructure, with Human Health accounting for 89.7% of the company's global sales as of the latest earnings report.

Within the Human Health division, Keytruda, Merck's flagship immuno-oncology treatment, remains the leading product and continues to experience robust global demand. It is complemented by a strong portfolio of vaccines, including Gardasil for the human papillomavirus (HPV), Vaxneuvance for pneumococcal disease, and several pediatric immunizations. The Human Health segment also includes therapies in diabetes, cardiovascular care, and infectious diseases. Though Keytruda represents

nearly half of Merck's total revenue, diversification is supported by other steadily performing drugs and emerging pipeline products.

Merck's geographic revenue distribution reflects a well-established international presence, though heavily concentrated in developed markets. According to data from 2022 to 2024 (see Figure 1), the United States accounted for 50.3% of total sales, confirming its position as Merck's most critical market. This is followed by Europe, the Middle East, and Africa (EMEA) at 21.8%, and Asia-Pacific and Latin America comprising the remainder. The steady year-over-year growth in the United States region highlights its strategic importance, while the slowdown in China during Q4 2024, especially affecting Gardasil, signals potential volatility in emerging markets. Despite this, untapped demand in under-immunized regions continues to support long-term international expansion potential.

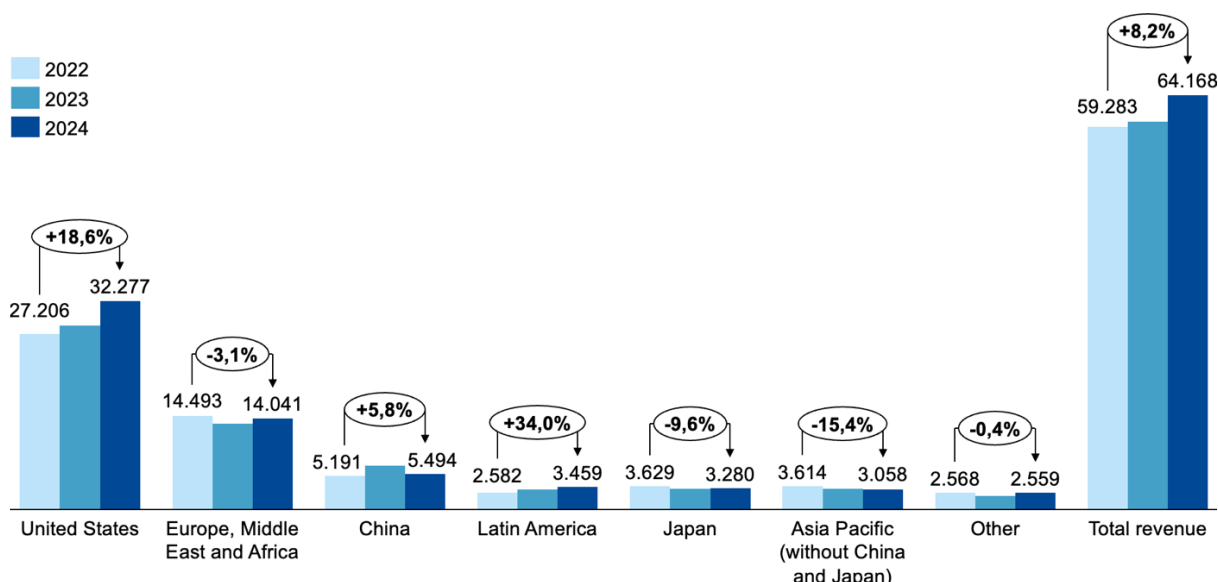


Figure 1: Revenues of Merck & Co by region from 2022-2024 in million USD (source: statista.com)

Operationally, Merck has a global footprint with strategic facilities positioned to support R&D, manufacturing, and regional market responsiveness. Its headquarters in Rahway, New Jersey anchors the business, while major international operations include biologics production in Ireland, research activities in London, and regional coordination and manufacturing in Singapore. Notably, the CFO has emphasized that Merck's manufacturing footprint is "slightly heavier outside the U.S.," which has implications considering shifting U.S. trade policy and potential future tariffs on pharmaceutical inputs.

2.3 Looking toward the future

Currently, Merck & Co are looking to broaden their product pipeline with 30+ new growth drivers that are currently undergoing the Phase 3 development program. These products can add further value to the company in the future if proven successful. In anticipation of global supply chain disruptions and rising demand, Merck recently announced the opening of a \$1 billion vaccine production facility and plans to invest \$8 billion in manufacturing and R&D expansion by 2028.

2.4 Current issues the business faces

Patent Expiry: A recent U.S securities filing earlier this year brings attention to the impact of the U.S. inflation reduction act on the key revenue driver of the company, namely, Keytruda. The product is targeted to have a “set price” by the first of January 2028, which is also when the patent expires. This effect is expected to depress sales revenue that the product generates, which currently makes up half of the company’s Total sales.

The CEO Rob Davis looks to resolve this problem by creating a subcutaneous version of the product which will be injected under the skin, rather than the current intravenous method involving administration into the veins. Administering into the skin allows greater access and maintenance over the alternative of IV placement, providing a competitive edge. This move is aimed to be complete by the first half of 2026, prior to the original version becoming a generic offering in 2028. With a generic alternative being introduced to the market in 2028, competition in this niche will increase. It is expected that Keytruda sales will see less growth from 2028 onwards however the new subcutaneous version Merck is working on will have a competitive advantage.

Slowing sales in China: While Merck’s vaccine segment remains a pillar of its Human Health business, recent data show signs of a short-term headwind in Gardasil sales, primarily driven by lower-than-expected demand in China. After nearly a decade of uninterrupted growth, during which Gardasil sales increased at a CAGR of 17.3%, the product experienced its first notable dip, declining from \$8.89 billion in 2023 to \$8.58 billion in 2024, a 3.4% year-over-year decrease (Figure 2). This slowdown is largely attributed to a sharp reduction in purchasing volumes in China, where demand had previously surged due to catch-up immunization programs and broader public health campaigns.

Despite this pullback, long-term fundamentals remain intact: HPV vaccination rates in many global markets, including China, remain significantly below WHO targets, suggesting ample headroom for future growth once market dynamics stabilize.

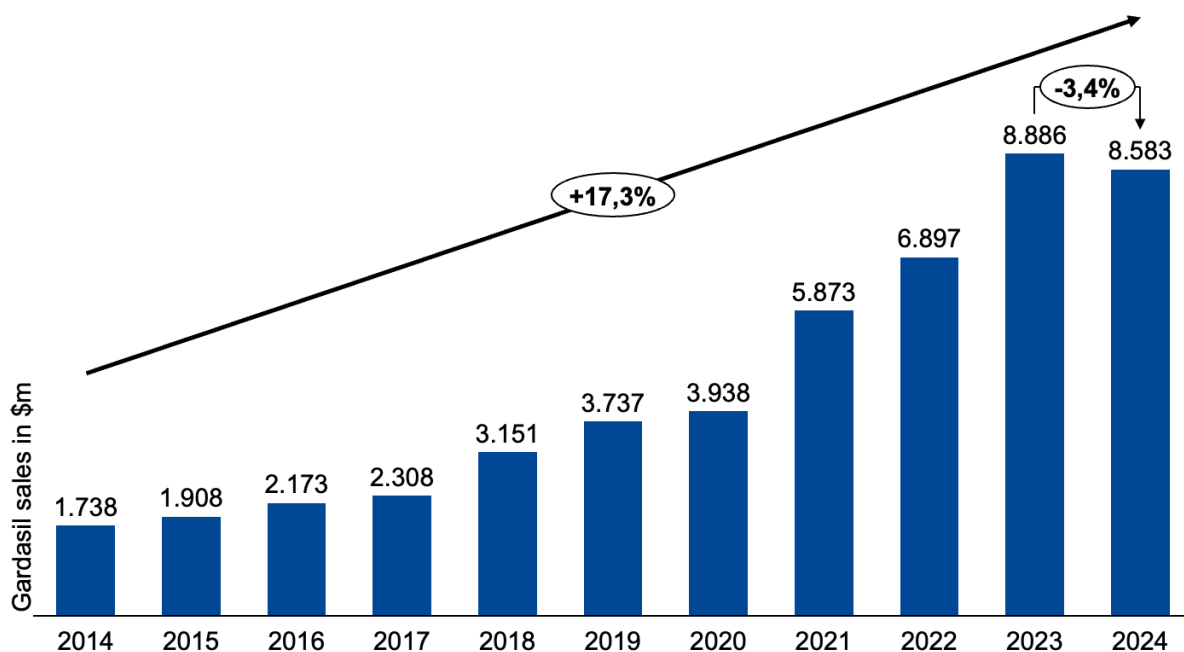


Figure 2: Gardasil sales year on year (source: Biospace)

Ongoing class action lawsuit: Currently there also exists a class action against the firm, stipulating securities fraud has occurred, with specific product sales in China not meeting forecasts, and this not being communicated to shareholders over an extended period. The lawsuit is a new initiative and seeks to claim compensation for investors who were adversely affected between February 2022 to February 2025. This is not a case that is expected to be resolved within the near term.

Tariffs: It is difficult to keep track of all the changes Trump is making to ongoing trade agreements on a global scale. As it stands Singapore, Ireland and England are part of the 90-day new Tariff pause. Currently the Tariffs do not cover pharmaceuticals, but importing medical devices, materials and ingredients will incur import taxes. The commerce secretary of U.S commerce has said “pharmaceutical tariffs are coming in the next month or two”.

It is evident that the Trump administration want to bring manufacturing back into America. Thus, it is positive to see that Merck & Co, had already planned to invest in

a new U.S manufacturing facility, prior to these tariff announcements. With a high Gross margin, the tariffs aren't expected to have major consequences, except for having some effects on the bottom line. It often seen that companies will pass these increased costs onto the customer. With pharmaceutical companies being non-cyclical and necessary, the products should always be in demand despite pricing levels, meaning it is anticipated that Merck & Co will pass the tariff taxes onto customers.

3. Industry Overview & Competitive Positioning

Merck & Co operates in several high-value pharmaceutical markets, with a strategic concentration in oncology and vaccines, two of the fastest-growing and scientifically dynamic therapeutic areas globally. These segments not only offer significant revenue opportunities but also feature high scientific complexity, creating substantial barriers to entry and reinforcing Merck's competitive edge.

3.1 *Market overview & Merck's Positioning*

Oncology: Oncology is the study and treatment of tumours. In 2024, the global size of this market was \$321 billion, and estimated to be \$356.2 billion in 2025. The global market is expected to grow at a CAGR of 10.9% per annum with the fastest growth seen in Europe at 15.8%. The United States is responsible for 47% of global revenue whereby 37% of this stems from lung cancer. Cancer is the leading cause of death globally, where lung cancer is responsible for the most deaths.

The WHO predicts a 77% increase in cases by 2050. The rapid rise in cancer is attributable to lifestyle changes, pollution, microplastics and an increasing geriatric population in developing countries. Additionally, the increasing challenges relating to increased cancer diagnosis and new forms of tumours means there is constant innovation and investment within the segment. This is evidenced by patents for cancer drugs are growing at a compounded rate of 9.4% per annum and research success rate rising to 10% in 2024, from 4% in the previous year.

The oncology market, driven by immunotherapies and biomarker-driven treatment approaches, is forecast to remain one of the most attractive areas in biopharma over the next decade. Merck's early leadership with Keytruda has cemented its position at

the forefront of this space. Meanwhile, the global vaccines market has expanded beyond traditional pediatric immunization into adolescent and adult prevention, pandemic preparedness, and emerging infectious diseases, areas where Merck has established strong foundations.

Vaccines: According to the WHO Global Vaccine Market Report 2024, the market is estimated at \$77 billion, a significant drop from the previous year due to decreased Covid-19 vaccine demand. While the 5-year CAGR from 2019 to 2023 is 3%, the 2023 CAGR spiked to 15% due to pandemic-driven demand, underscoring the impact of global health crises on vaccine consumption.

Despite the decline, the vaccine market remains a vital, resilient segment of the pharmaceutical industry, driven by strong public-sector demand, high entry barriers, and long product cycles. Merck stands out, particularly in HPV and pneumococcal vaccines, with strong global franchises.

Routine immunization demand has stabilized at elevated levels post-pandemic, supported by improved healthcare access in emerging markets and greater government investment in disease prevention. However, regional volatility persists due to procurement cycles and pricing dynamics.

Merck's Gardasil leads in adolescent and adult immunizations, though 2024 saw a 3.4% decline in sales (Figure 2), mainly due to reduced Chinese demand, a short-term correction rather than a lasting trend. In pneumococcal vaccines, Merck's Vaxneuvance is gaining ground on Pfizer's Prevnar, praised for broader serotype coverage.

Merck & Co also maintains a strong position in pediatric combination vaccines, ensuring steady revenue and reinforcing its role as a key public health partner.

Overall, Merck is well-positioned in the vaccines market through a mix of innovation, established global partnerships, and operational scale. The company's ability to adapt to regional shifts, like the recent slowdown in China, while continuing to invest in next-gen vaccines, underpins its long-term competitiveness in this essential segment of healthcare.

Animal health: The global Animal health market was valued at \$62.89 billion in 2024. The United States boasts the largest revenue share, by geographic area, equal to 35.7%. The industry is set to grow at a CAGR of 10.46%, attributable to Artificial

Intelligence, increasing disease in animals, increased pet insurance and ownership and finally pet humanization.

Defining the roll of AI, is the increased roll of AI being used to develop drugs for animal treatments. The increased humanization of pets' references to pet owners caring and catering to their pets, more than the average pet owner, increasing insurance and healthcare treatment. Companies operating within this space tend to specialise in a niche. Product insights show that the pharmaceutical segment makes up about 43% of sales and includes drugs for anti-inflammation, parasites and antibiotics.

Apart from the category of companion pets, there also exists a production animal segment, animals that are bred for production and consumption. An increased interest in food safety and sustainability has led to increased healthcare costs pertaining to this segment. These factors are all expected to benefit the market, contributing to the positive CAGR. Overall, the Animal health sector is a highly competitive market, owing to the existence of many companies. In such a market, competitive advantage, presence and product penetration are important to be successful.

Although smaller in absolute terms compared to Merck's Human Health segment, Animal Health contributes to earnings and provides a valuable hedge against volatility in other parts of the business. With growing demand across both companion and production animals, and room for continued expansion in diagnostics and digital tools, this division is expected to remain a durable and increasingly strategic asset in Merck's portfolio.

Summary: Combining the segments and defining them as being part of the pharmaceutical industry, long term growth out to 2032 for the pharma industry is quoted to be 6.4%, according to research released by Acumen consulting. This is driven by an aging population, and increasing chronic diseases, advances in medicine and rising healthcare awareness.

3.2 Competitive positioning

The global pharmaceutical market is estimated to be worth \$1.6 trillion in 2024. Within this space operate some large household name companies that are constantly developing new treatments, contributing to the high growth of the industry. Ranking the major players in this industry by their share of sales revenue in 2024, Pfizer comes in

at number one. Pfizer is followed by Johnson & Johnson, AbbVie, Merck & Co and finally Roche respectively.

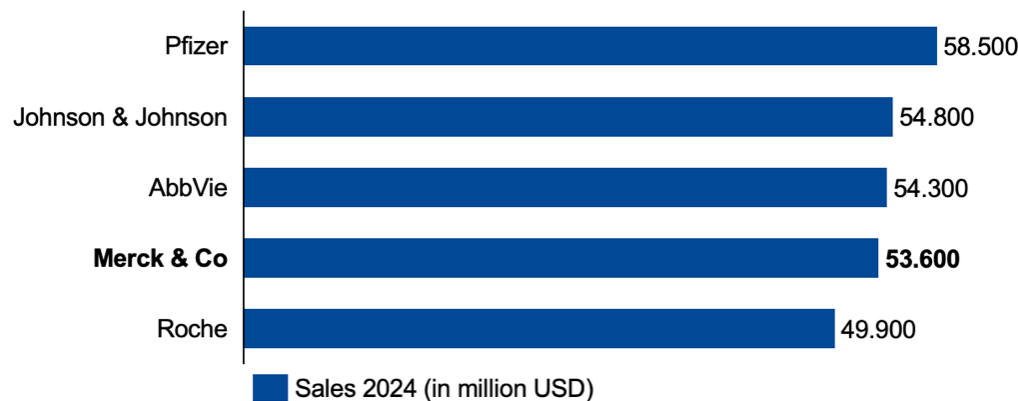


Figure 3: Sales revenue comparison 2024 of large players in the pharmaceutical industry (source: own illustration)

Figure 3 illustrates that Pfizer remains the largest pharmaceutical company by sales, largely due to its scale, broader therapeutic diversification, and revenue from vaccine and treatment products. Johnson & Johnson follows, benefiting from its diversified structure that includes pharmaceuticals, medical devices, and consumer health. AbbVie, driven by the success of Humira and its follow-on assets, narrowly edges out Merck. Roche, traditionally a powerhouse in oncology, ranks just below Merck & Co as its revenue growth slows due to biosimilar erosion in legacy cancer drugs.

Whilst these dominant pharmaceutical companies operate within the same industry, each company focuses on niches and thus it is not necessarily true that they are direct competitors across all product ranges, due to the specialisations and focus lying in different segments, differentiating them from each other.

The cluster diagram in Figure 4 shows that Merck & Co stands out from its competitors due to its specialization in animal health. This is a business segment that none of the other companies in the diagram cover, giving Merck & Co a unique market position. Compared to Pfizer and Roche, which have a particularly broad diversification across multiple therapeutic areas, Merck & Co's portfolio is relatively more limited.

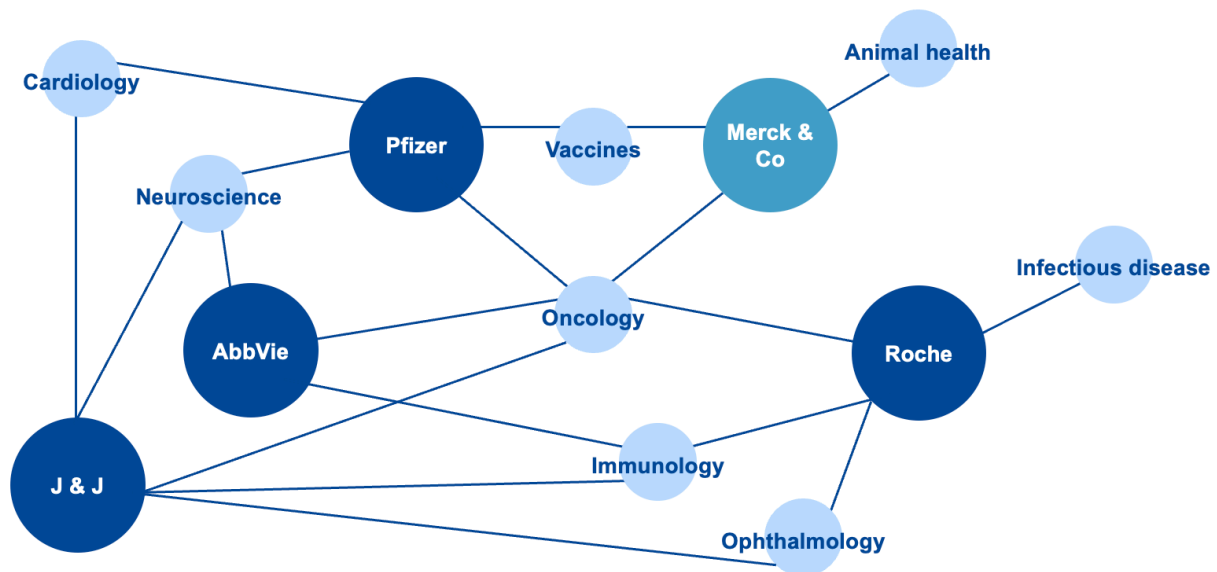


Figure 4: Cluster diagram of the market segments of Merck & Co and its competitors (source: own illustration)

While companies like Roche have additional specializations in areas such as infectious disease and ophthalmology, Merck & Co remains primarily active in traditional pharmaceutical markets such as oncology and vaccines. However, the company is not present in other key segments like neuroscience or ophthalmology.

Analysing Merck & Co's distribution of global sales, the reliance on sales from the United States of America is amicable. In the future it would be beneficial to see greater diversity in this aspect. The Japanese market should be an area of interest with the Oncology market expected to reach nearly \$30 billion by 2030, reflecting a CAGR of 12.3%. Japan is the 3rd largest economy in the world, however, is special because of their strong dependence of culture, values and "made in Japan" products.

The market is also protected by many regulatory hurdles, making it difficult for foreign companies to enter. The high barrier for entry means there would be fewer competitors. Seeing as Merck & Co already generate sales in this region, it could be an area of focus for future products and growth. It is also expected that China sales will continue to be subdued following a fall in demand for the firm's second most sold product, Gardasil, a vaccine fighting HPV. This slump in sales across Q3 and Q4 of 2024, resulted in the company missing revenue forecasts, strengthening the argument that the company needs a stronger product portfolio.

In summary, while Merck & Co may not be the largest player by revenue, its strategic focus, product leadership, and R&D pipeline strength secure its place as a top-tier

competitor in the global pharmaceutical landscape. Its ability to innovate in oncology, vaccines, and animal health allows it to hold ground, and in many cases, lead against more diversified peers, reinforcing its role as a high-performing, innovation-driven company in the sector.

3.3 Porter's five forces

To really understand where Merck & Co stands in the pharmaceutical world, it helps to look at the bigger picture, what kind of pressure they're dealing with from competitors, new players, customers, and suppliers. Porter's Five Forces is a useful tool for this. It breaks things down into five key areas that affect how tough the market is and what kind of edge a company might have. Here's how those forces play out for Merck.

3.3.1 Competitive rivalry

- Merck competes with big names like Pfizer, Roche, Johnson & Johnson and AbbVie. These companies are constantly developing new drugs, especially in across all indications. The continuous competition means Merck & Co must keep investing in research and innovation to maintain its position.
- Additionally, Merck's top competitors like Johnson & Johnson and Roche have much broader product portfolios. Compared to its competitors, Merck's focus is more specialized, mainly around oncology, vaccines, and animal health, which means it needs to stay highly competitive in these core areas to maintain its market share.

3.3.2 Threat of new entrants

- Starting a pharmaceutical company has high barriers to enter. It takes a lot of money, time, trials and approvals to bring a drug to market. Because of these barriers, it's unlikely that new companies will enter and have an immediate effect against established players in the pharmaceutical industry.

3.3.3 Supplier bargaining power

- Merck works with different suppliers across the world. While it has options, things like import taxes and supply chain issues can still impact operations. The company's size gives it some leverage, but it's not completely immune to global trade changes. Geopolitical issues like tariffs and material costs can affect its

supply chain. Still, the company's size and global operations offer some negotiation power.

3.3.4 Customer bargaining power

- Because healthcare is a non-cyclical industry, demand for Merck's products stays strong regardless of the economy. This limits the power buyers have, especially when it comes to essential or patented drugs. However, in cases where alternative or generic versions exist, buyers like hospitals and governments may use that to negotiate lower prices. So, while buyer power is generally low, it increases when substitutes are available.

3.3.5 Substitutes

- Once a drug's patent ends, cheaper generic versions often hit the market. This is a major challenge for Merck, especially with its top-selling drug, Keytruda, losing patent protection in 2028. Generic competition can steal market share as an affordable alternative.

3.4 SWOT-Analysis

The Strengths, Weaknesses, Opportunities and Threats analysis infographic (Figure 5) brings further attention to the company's competitive positioning.

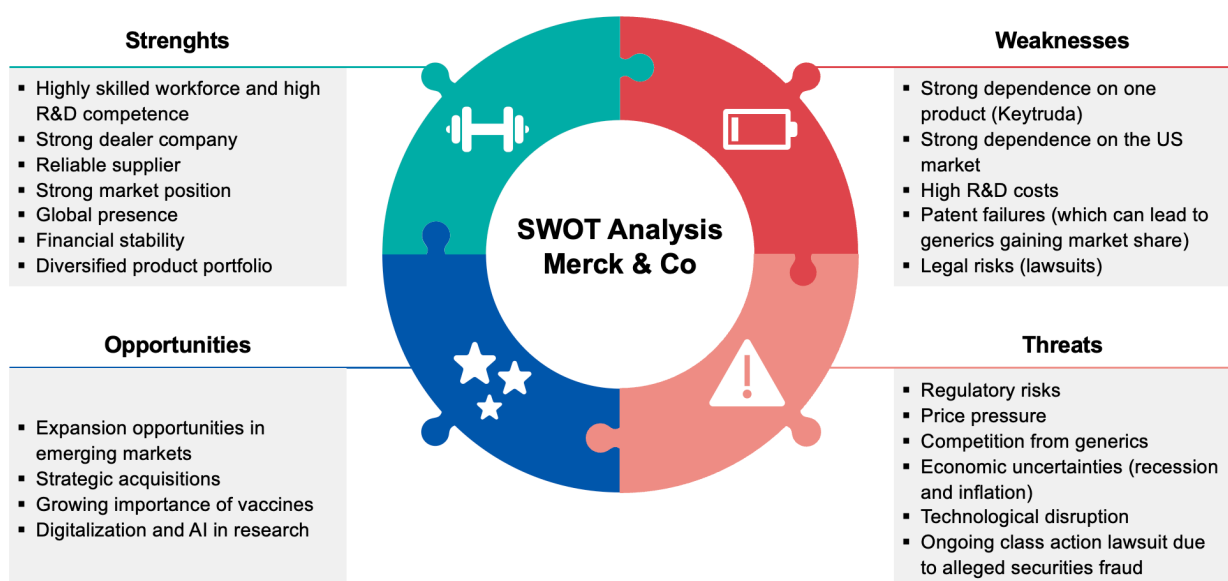


Figure 5: SWOT Analysis of Merck & Co (source: own illustration)

The SWOT analysis illustrates that Merck's core strength lies in deep scientific capability, operational efficiency, and product leadership. However, the company must actively de-risk its reliance on Keytruda, accelerate diversification into adjacent therapeutic areas, and maintain proactive geopolitical and regulatory navigation. With a strong pipeline, scalable global infrastructure, and strategic investments in R&D and manufacturing, Merck is well-positioned to turn many of today's risks into tomorrow's opportunities.

4. Corporate Governance & Share Ownership

Examining the board structure of the firm, the company is led by Chairman and CEO Robert Davis. Rob has progressed through the firm, earlier holding titles of president and vice president and chief financial officer. Rob was earlier president for another healthcare company before moving to Merck & Co in 2014.

Assisting Rob, there is a large team of competent leaders that have held earlier job roles at Wyeth, Johnson & Johnson, and Baxter- Which are all large healthcare firms. The average tenure of management is 4.3 years, with the board of directors' experience exceeding this at 5.6 years on average. This indicates that the leaders of the firm are an experienced team and know the company at a deeper level.

Analysing Insider trading, in the last three months, the shares sold vs shares bought is 16,861 against 17,832 respectively. It is seen as a positive that more shares have been purchased, although a higher buy-to-sell ratio would be preferred.

It is also relevant to mention management hold only a small stake in the company at 0.07%. 80% of shares are held by institutions which could affect liquidity, dependent on the institutions' strategies and holding period.

5. Fundamental Analysis

Merck & Co is a stalwart, that continues to be reliable and stable. The experienced management are trusted to make the correct decisions for the future direction of the company, with increasing focus on a broader product range, supported by 30+

products currently in stage three trials. From the Financial Year 2024 annual report, key findings were:

- Revenue growth of 7% year on year
- Keytruda sales increased by 21% whilst animal health grew at 9%
- Gardasil vaccine sales impact in China not meeting guidance
- Company is confident in navigating Keytruda patent expiry
- Gross margin for Q4 2024 was 80.8%, up 3.6pts from same time last year, this is expected to increase to 82.5% in 2025.

	Merck & Co	Pfizer	J & J	AbbVie	Roche	Ind Avg.
Gross Margin	77.1%	74.2%	69.6%	70.3%	74.8%	69.3%
ROE	36.9%	9.1%	20.1%	62.3%	26.5%	32.1%
ROA	14.6%	4.6%	7.8%	8.3%	13.6%	9.5%
Asset turnover	54.8%	30.0%	51.0%	42.0%	65.0%	48.0%
Current ratio	1.36	1.17	1.11	0.66	1.48	1.17
Debt/equity	84.6%	76.2%	52.9%	2.0%	100.6%	84.2%
P/B	4.4	1.7	5.6	108.9	7.3	-54.4
P/E	11.95	19.0	28.7	89.3	31.1	26.8

Table 1: Financial ratios of Merck & Co, its competitors and the industry average (source: investing.com)

The strengths of the company are featured in the table above (Table 1), when comparing key financial ratios against peers and the industry average.

Analysing the financial ratios in this table, it is evident that Merck & Co is in a solid position compared to its main competitors but also outperforms the industry average.

5.1 Gross margin

The gross margin is best in class, indicating that Merck & Co keeps 77.1% of revenue after deducting cost of goods sold. Merck & Co have revealed that the gross margin is forecast to equal 82.5% in 2025. This will position the company ahead of competitors and the industry average, providing more leniency in pricing strategies.

5.2 Return on Equity

The ROE is an indicator for firm performance and the returns that the company is earning on invested capital. Merck & Co has a ROE higher than the industry average,

meaning they are utilising their equity better than the average competitor, to generate positive returns.

5.3 Return on Assets

A strong return on asset figure of 14.6%, reveals that the company is above the industry mean when measuring profits against assets. This indicates that the company is using its assets much more effectively than the industry average, to produce sales.

5.4 Asset turnover

This ratio reveals how efficiently a company is using its assets to generate sales. Merck & Co holds the second highest rate, and are above the industry average, showcasing that the company is using their assets more effectively than their competitors.

5.5 Current ratio

Looking at the liquidity of the business, the current ratio shows that the company can meet short-term obligations using short term assets. This ratio exceeds that of competitors revealing that Merck & Co are more liquid.

5.6 Debt to equity

This ratio shows the amount of debt, relative to equity. Merck & Co's ratio of 84.6% indicates that the firm has undertaken debt to finance investments. A figure less than 100% is considered good, where a higher ratio will make it more difficult to borrow money. Merck & Co is sitting at the industry average, not deviating from the norm in this respect.

5.7 Price to book

the price-to-book ratio shows that Merck & Co is significantly higher than the industry average, however, does not satisfy the value proposition, where a P/B ratio less than 1.0 is defined as undervalued. The ratio indicates that the firm is valued at a multiple of 4.4 its book value of equity. A negative P/B value would suggest that liabilities exceed assets, which could indicate a poor investment.

5.8 Price to earnings

This ratio shows that the industry tends to trade at a multiple of 26.8 times earnings. In this aspect Merck & Co is trading below par. Despite that, the P/E is subjective and based on risk factors and growth, whereby higher growth rates increase the stock price whilst risk will reduce stock prices. In respect to the industry average the company is undervalued.

5.9 Summary

In conclusion, the ratios reveal that Merck & Co are overall performing better than their peers and the industry average. This is supported by greater gross margins, efficient use of assets to generate profits and limiting the use of debt for investment purposes. These views are shown by the valuation ratios, where the P/B ratio shows there is potential value, coupled with the firm trading at a P/E multiple, well below the industry mean and competitors. Hence, the ratios underline that Merck & Co are a solid, well-established company that is undervalued when compared to peers.

6. Investment Risks

Investments are subject to various risks that may affect the expected return, and the safety of the capital invested. Investment risks can be divided into several categories, each of which has a different impact. In the following, the risks of Merck & Co have been extracted from the annual report and divided into the following categories: Market Risk (MR), Operational Risk (OR), Financial Risk (FR), Legal and Regulatory Risk (LR). Each risk was given a name for ease of identification, and the risks were then categorised in an impact-probability matrix (see Figure 6).

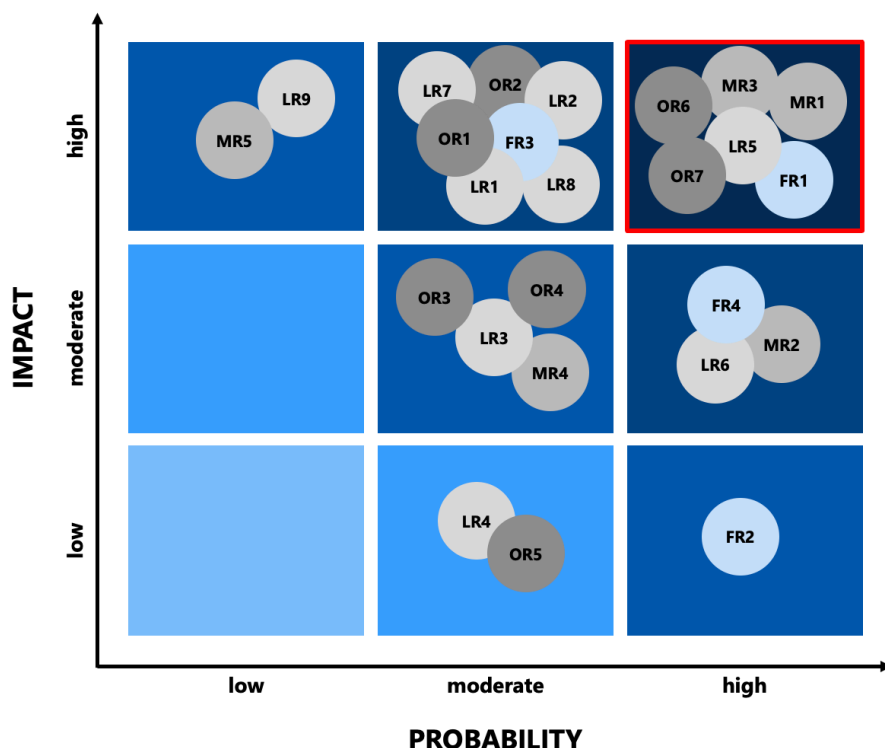


Figure 6: Impact-probability matrix (source: own illustration; based on annual report)

Of the 25 risks identified, six have both a high probability of occurrence and a significant impact. These particularly critical risks are analysed in detail below. Not only is their potential impact on the company analysed, but appropriate mitigation measures are also identified to limit their effects and ensure sustainable risk management.

6.1 Market Risk (MR)

MR1 - Loss of market exclusivity and its effects: As Merck & Co's products lose market exclusivity, the Company generally experiences a significant and rapid loss of sales from those products. This is because, following the loss of exclusivity, generic alternatives or copycat products are entering the market, which are often offered at significantly lower prices.

Mitigation: To counteract the negative effects of market exclusivity, Merck & Co can rely on continuous investment in R&D to develop new products and thus tap into new sources of sales, thereby compensating for the decline in sales of existing products. In addition, Merck & Co can continue to successfully position its products even after the market entry of generics through a targeted brand strategy and by building trust and customer loyalty.

MR3 - Intense competition from generics and competing products: The intensity of competition from low-priced generics as well as from innovative competing products can have a negative impact on market share, sales and profitability.

Mitigation: In addition to the proposals from mitigation in risk MR1, Merck & Co can achieve a reduction in costs by increasing efficiency in products and sales and offset sales losses through strategic partnerships and the development of new markets. In addition, targeted market segmentation in price-sensitive markets can increase competitiveness.

6.2 Operational Risk (OR)

OR6 - Risks and uncertainties in the field of biologics and vaccines: Due to their complexity, biologics and vaccines carry unique risks and uncertainties that could have a material adverse effect on the Merck & Co's future results of operations and financial condition.

Mitigation: Production risks can be minimised by investing in modern biotechnology, automation and quality control systems. Diversification of product locations can help to avoid supply chain disruptions. Cooperations with research institutions or other biotech companies can help to share development costs and spread risks.

OR7 - The need for authorisation and its maintenance: Merck & Co's products, including products in development, cannot be marketed unless the Company obtains and maintains regulatory approval.

Mitigation: Proactive collaboration with regulatory authorities and local health authorities can help to understand regulatory requirements at an early stage and identify potential hurdles. In addition, careful planning of clinical trials can reduce the risk of delays or insufficient trial data.

6.3 Financial Risk (FR)

FR1 - Dependence on key products and their impact: Key products generate a significant amount of the Merck & Co's profits and cash flows, and any events that adversely affect the markets for its leading products could have a material adverse effect on the Company's results of operations and financial condition. This risk is apparent with the flagship product Keytruda being price set in 2028, which will have adverse effects on sales revenue.

Mitigation: A strategic expansion of the product portfolio through in-house R&D and through acquisitions and partnerships can contribute to a broadly diversified risk. Market exclusivity can also be extended through patent extensions or the introduction of new identification variants of existing products. This enables Merck & Co to minimise the risks of a sudden drop in sales and maintain a stable financial basis in the long term.

6.4 Legal and Regulatory Risk (LR)

LR5 - Risks from regulation and political influence in the healthcare sector: The health care industry has been, and will continue to be, subject to increasing regulation and political action.

Mitigation: Political lobbying and early adaptation to new regulatory requirements can help to minimise the risk. In addition, expanding into less regulated markets or developing products that are accepted in different healthcare systems can spread the risk.

6.5 Risks Summary

Summarized, Merck & Co faces key risks across market, operational, financial, and regulatory domains. Market risk includes the loss of exclusivity (MR1), leading to rapid sales decline due to cheaper generics. Merck mitigates this through ongoing R&D, strong branding, and customer loyalty efforts. The company also contends with intense competition (MR3) from generics and innovative products, addressed by improving efficiency, forming partnerships, and targeting new or price-sensitive markets.

Operational risks involve the complexity of biologics and vaccines (OR6), which carry production and quality challenges. Merck responds with investments in biotech, automation, quality control, and diversified manufacturing. Regulatory approval (OR7) is another critical risk, mitigated through early engagement with authorities and strategic clinical trial planning.

On the financial side, Merck is heavily reliant on key products like Keytruda (FR1), which face pricing pressures in coming years. The company counters this by broadening its portfolio via R&D, acquisitions, and patent strategies to maintain revenue stability.

Finally, regulatory and political risks (LR5) are managed through proactive lobbying, early adaptation to new rules, and market diversification to reduce exposure to any one system or policy shift.

7. Environmental, Social and Governance (ESG)

ESG criteria are of significance to investors as they minimise risk and promote sustainable returns. Companies with a strong ESG profile are more resilient, innovative and attractive to investors. In addition, a good ESG strategy facilitates access to financing and strengthens competitiveness. As regulatory requirements and demand for sustainability increase, ESG criteria are a key factor for long-term economic success.

According to the S&P Global ESG Score, Merck & Co has a score of 52 out of 100, which is in the middle range. Compared to the industry average of 35, the company has a higher score, especially in the field of corporate governance (see Figure 7). The rating agency Sustainalytics also gives Merck & Co a medium risk rating, comparable to other market participants.

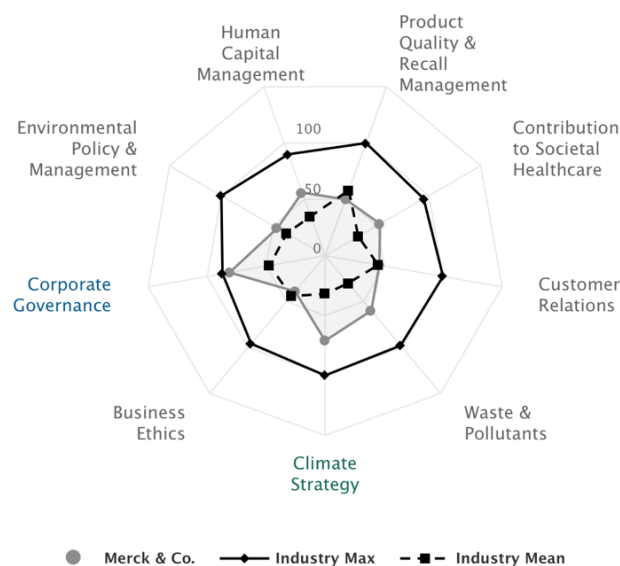


Figure 7: Company vs Industry Performance (source: S&P Global)

The latest ESG progress report shows that Merck & Co faces challenges in achieving its ambitious decarbonisation targets of becoming climate neutral by 2025. Although Scope 1 and Scope 2 emissions have been reduced slightly compared to the baseline in 2019, while Scope 3 emissions have increased by 9 %. The strategy to achieve climate neutrality, which requires electricity to be sourced entirely from renewable

sources, has also not yet been fully implemented. By 2021, only 41% of this target had been achieved, which shows that Merck & Co still must make significant efforts to realise the climate targets set on time and achieve a significant reduction in emissions in all areas.

As a pharmaceutical company, Merck & Co sees the third Sustainable Development Goals of the United Nations (Good Health and Well-being) as core to its business and recognises its responsibility to improve access to medicines and vaccines around the world. In addition to these SDG's, the company is focussing on seven other goals where their greatest impact is expected.



Figure 8: SDGs of Merck & Co (source: ESG progress report 2021/2022)

Merck & Co is committed to establishing further social standards by auditing its suppliers about labor rights, environmental protection and ethical standards. The company is also a member of the Pharmaceutical Supply Chain Initiative (PSCI), which promotes sustainable practices within the supply chain and thus contributes to improving social and environmental responsibility throughout the industry.

The company has a strong governance structure based on ethical business practices and close stakeholder engagement. The Board meets at least six times a year to make strategic decisions and review progress in various areas.

Merck & Co has established four committees (Audit, Compensation & Management Development, Governance and Research) consisting exclusively of independent directors. Among other things, these committees monitor ESG matters and ensure that relevant ESG issues are reviewed by the relevant committees.

Executive compensation at Merck & Co is based on performance, with bonuses linked to financial, operational and ESG targets. The company also engages in dialogue with its shareholders to understand their perspectives, particularly on governance, compensation and ESG issues. This proactive shareholder engagement program and regular discussions provide the basis for informed decisions that are made in the best interests of the company and its shareholders.

8. Valuation

Within the next section, two separate equity valuation techniques will be conducted. The average of the two derived valuations will be used as the true valuation and target price. A dividend discount model (DDM) and a free cash flow to equity model (FCFE) will each be created. Across both models, the same discount rate and perpetual growth rate is used, to maintain consistency throughout.

8.1 Defining the discount rate and perpetual growth rates (CAPM)

The Capital asset pricing model will be used for the discount rate as it reflects the expected returns of equity holders. Using the calculated beta of 0.455, average SPY 500 return 10% and the risk-free rate of 4.34%, the expected return of an investor is 6.91%. The beta indicates that there is less volatility than the market, thus investors require less than market return, because there is less risk. A conservative perpetual growth rate of 4% per year was allocated based on the expected long-term growth trends that were researched by Acumen consulting. Reasoning for being conservative is that the pharmaceutical industry is a broad umbrella and Merck & Co, are focused on specific indications within this sector. Additionally, uncertainty in the global markets now contributes to choosing a lower perpetual growth rate (Figure 13).

8.2 Assumptions for future cash flows (base case)

Developing a base case scenario to capture the fast-growing pharmaceutical market and future product line of Merck & Co, the sales revenue is an important aspect to which many subcategories cashflows are based off.

Breaking sales up into key components, there are two categories. Human health and animal health. Keytruda has been extracted outside of human health for the reason of it being 50% of sales and exhibiting fast growth, year-on-year. Projecting human health excluding Keytruda, a flat profile of 7% growth p/a was applied. Animal health exhibits faster growth in financial reports and has been dedicated a flat growth profile of 8% p/a. With Keytruda recording an average CAGR of 21% across the past three years, we expect sales in the following years to be similar.

However, Keytruda has implications in 2028 when the patent expires, which is expected to hamper sales due to generic options becoming available. For this reason,

the product has been dealt a three-stage growth model; 20% in 2026, 15% in 2027 and 10% from 2028 to 2030. This provides a CAGR of 12.93% for the product (Figure 14).

Other aspects and assumptions of the model were provided by Merck & Co as guidance or have been estimated based on the three-year historical average, as a percentage of revenue for each projected year. The financial reports indicated gross margins were expected to increase to 82.5%, and the corporate tax rate is equal to 16.5%. Further guidance for the 2025 financial year, Merck & Co expect revenue to grow between 0 to 2%. It was also revealed that the company expects to invest an additional \$4 billion in research facilities in the years 2026 and 2027, which has been incorporated into the FCFE model, as an addition to the normal CAPEX expenditure.

The valuation of Merck & Co based on these projections and assumptions (Figure 15) is calculated as being \$126.93 per share. This reflects a premium to today's share price, signalling the company is currently undervalued.

8.3 Assumptions for future cash flows (bull case)

The presented bull case scenario for Merck & Co outlines an optimistic view of the company's future development. This valuation scenario is primarily driven by strong expected revenue growth across its key segments, particularly from its blockbuster drug Keytruda, the human health division, and the animal health segment.

The most important growth driver in this scenario is Keytruda, which is forecasted to grow at an average annual rate (CAGR) of approximately 18% through 2030 (Figure 16). This growth is staged with 20% sales growth up 2028. For 2029 and 2030 the sales growth is reduced to 15%, reflecting the generic products coming to market and stealing market share. Given the drug's current market penetration, future indications, and Merck's robust pipeline, this growth assumption is ambitious yet plausible from a bullish perspective. At the same time, the remaining human health portfolio is projected to grow at around 9% annually, while the animal health segment is expected to contribute with solid growth of 10% year on year.

As a result, total company revenues are projected to grow at a strong CAGR of 13.08% from 2025 to 2030. Sales are expected to increase from approximately \$63 billion in 2025 to nearly \$117 billion in 2030 representing almost a doubling of revenues within six years. This underlines the dynamic growth assumption underlying this bull case.

On the earnings side, the company is expected to return to sustainable profitability starting from 2025 after a weaker period in 2023. EBIT is forecasted to nearly double from around \$12 billion in 2025 to just over \$22 billion by 2030. This development reflects both scaling effects from rising sales volumes and operational efficiency improvements. Despite continuous investment in R&D and administrative expenses, the cost base remains well controlled in this scenario.

A key metric for valuation purposes is the development of FCFE. After a weak period in 2023 and 2025, FCFE is projected to grow significantly from 2026 onwards, reaching over \$13 billion annually by 2030. This strong cash flow generation demonstrates Merck's ability to deliver substantial returns to shareholders, despite its heavy investments in innovation and expansion.

Based on these projected cash flows, the net present value of FCFE is estimated at \$365 billion in the bull case. This results in a calculated fair value per share of \$144.56 (Figure 17), significantly above Merck's current share price, indicating considerable upside potential if these projections materialize.

Another positive takeaway is the consistent increase in earnings per share, which is projected to rise from \$4.74 in 2025 to \$8.77 in 2030. This reflects both the company's improving profitability and the efficiency of its capital allocation strategy.

In summary, this bull case scenario highlights Merck & Co.'s strong market position and excellent growth prospects, largely driven by Keytruda and additional pipeline opportunities. The company demonstrates robust revenue dynamics, stable margins, and strong cash flow generation capabilities. The resulting fair value estimate of \$144.56 per share indicates substantial upside potential for investors who believe in the realization of this optimistic scenario.

8.4 Assumptions for future cash flows (bear case)

The presented bear case scenario for Merck & Co reflects a rather conservative view on the company's future performance, considering moderate sales growth, and the dependency on key growth drivers such as Keytruda delivering sub optimal revenue figures.

Sales projections indicate a CAGR of 7.46% for total revenue between 2025 and 2030. This is driven primarily by the oncology drug Keytruda, which is expected to grow at a

strong CAGR of 9.87%. While human health sales and animal health are expected to grow at flat a flat 5% and 6% per annum respectively (Figure 18). Overall, this growth outlook supports a stable but unspectacular top-line sales development in the bear case.

From 2025 onwards, profitability increases in line with the slow growth in sales. With the company having 82.5% gross margins, the company has a lot of leeway in respect to expenses and capital expenditure. From 2025 to 2030, the EBIT rises year on year, having grown 43.3% over the 5-year period. The FCFE follows a similar pattern, reaching, reaching approximately \$10.2 billion by 2030.

Coming up with a valuation based on this scenario, the derived net present value of free cash flows to equity holders is circa \$287 billion, with a terminal value of \$349.06 billion. Discounted back to May 2025, the total enterprise value leads to a fair stock price of approximately \$113.74 per share. This valuation reflects a cautious but constructive view on Merck's long-term fundamentals, if the company's sales are impacted by macroeconomic events.

In summary, the bear case scenario illustrates that Merck & Co can operate on less-than-ideal sales, cushioned by their large gross margins. However, whilst being able to stay cashflow positive, the earnings per share will be impacted. In years of lower revenue, the company will need to be observant of ensuring their capital expenditure and admin costs do not lead to a loss-making year. The historical moderate but stable sales growth across all business segments and the recovery of free cash flows support the view that downside risk for the stock is relatively limited, assuming no major pipeline failures or regulatory setbacks.

Overall, the bear case fair value of \$113.74 (Figure 19) per share provides a useful benchmark for investors, highlighting that even in a more cautious scenario, Merck & Co's stock offers value compared to today's share price. This resilience and value stems from its diversified business model, high growth margins and future product pipeline that can contribute to future sales.

8.5 Dividend Discount Model

The Dividend Discount Model (DDM) is a fundamental valuation method that estimates the intrinsic value of a company's equity by projecting its expected future dividends

and discounting them back to present value. This model is particularly suitable for firms with a consistent dividend payment history and a stable outlook for long-term cash flows, conditions that apply well to Merck & Co.

Merck has maintained a reliable dividend policy for many years, reflecting both its strong cash generation and management's commitment to shareholder returns. The company has increased its dividend consistently over the past decade, making it a suitable candidate for the DDM approach. While this model may not capture the full complexity of a large R&D-driven business like Merck, it provides a useful cross-check alongside the FCFE method.

Between 2019 and 2024, Merck's dividends per share grew at a compound CAGR of approximately 6.67%. This sustained growth has been supported by strong earnings, margin expansion, and disciplined capital allocation.

The model assumes that this 6.67% annual growth rate will continue through to 2030, which aligns with historical trends and management's implicit signalling through guidance and capital expenditure plans. Given Merck's ongoing investments in pipeline assets and capacity expansion, this growth rate is considered moderate yet sustainable.

Based on these calculations, the DDM yields a fair value of \$105.22 per share (Figure 20). This is below the FCFE base case valuation of \$126.93 but still above the current market price, supporting the view that Merck is undervalued from a dividend yield perspective.

The DDM provides a conservative valuation baseline, as it only considers cash returned to shareholders and excludes the value of retained earnings that are reinvested into growth. It is sensitive to small changes in discount rate or growth assumptions, making it best used as a complement to free cash flow-based models.

That said, the fact that the DDM result remains significantly above the current market price reinforces Merck's value proposition to income-focused investors, particularly given the company's consistent dividend track record, defensive business model, and commitment to long-term shareholder returns.

8.6 Valuation summary

To estimate the intrinsic value of Merck & Co., two complementary valuation models were employed: the FCFE model and the DDM. These approaches provide different but reinforcing perspectives on Merck's equity value. The FCFE model incorporates detailed operational forecasts and capital allocation assumptions, making it more dynamic and forward-looking. The DDM, on the other hand, offers a simpler, more conservative view focused solely on shareholder payouts.

In the base case scenario, the FCFE model estimates Merck's intrinsic value at \$126.93 per share, based on a realistic revenue growth path, continued margin improvement, and execution of planned capital investments. This scenario assumes that the company successfully manages the transition around Keytruda's 2028 patent expiry, mitigating the impact with a subcutaneous version and pipeline replacements.

The bull case yields a valuation of \$144.56, reflecting stronger-than-expected revenue growth, robust adoption of new therapies, and efficient operational scaling. Conversely, the bear case, with more cautious growth assumptions and potential headwinds from generic competition and regional volatility, still supports a valuation of \$113.74, well above Merck's current market price. This shows a limited downside risk, even in more challenging conditions.

To complement the FCFE valuation, a Dividend Discount Model was also constructed. Using a 6.67% historical dividend growth rate and a perpetual growth assumption of 5%, the DDM produces a valuation of \$105.22 per share. While lower than the FCFE estimate, it remains above Merck's market price, underscoring the stock's appeal to dividend-focused investors and the sustainability of its payout policy.

These results are summarized in Figure 9, which clearly presents the valuation across various models and scenarios. The graphic illustrates how Merck's intrinsic value consistently exceeds its current market price under all modelled conditions.

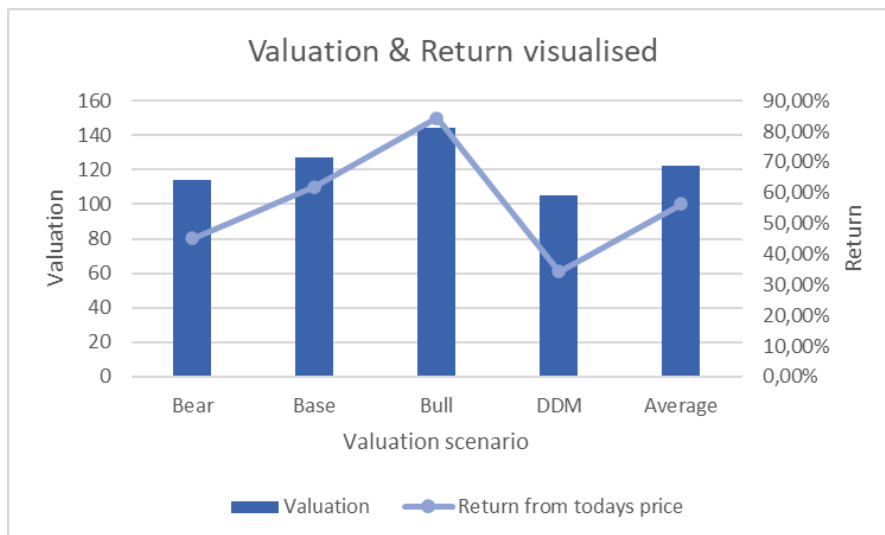


Figure 9: Valuation & return visualised (source: own illustration)

Further insights are provided in Figure 10, which visualizes the valuation range across the bear, base, and bull case scenarios, as well as EPS development or share price sensitivity over time. The figure highlights Merck's upward earnings trajectory through 2030 and provides a visual confirmation of how increasing revenue and operational leverage contribute to rising intrinsic value. The consistent increase in earnings per share, even in more conservative cases, demonstrates Merck's ability to generate shareholder value despite near-term pressures, such as the Keytruda patent expiry or temporary sales slowdowns in China.

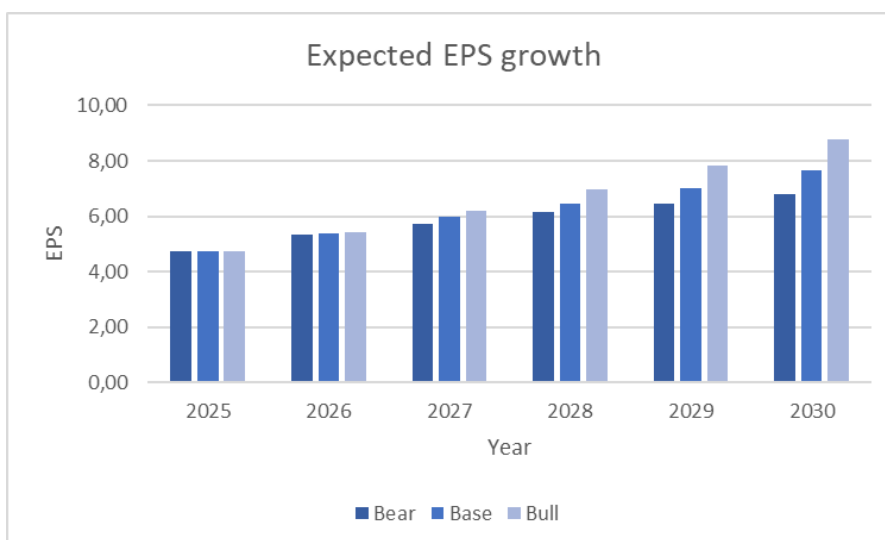


Figure 10: Expected EPS growth (source: own illustration)

Taken together, these models and visual summaries strongly reinforce the investment thesis: Merck is currently undervalued, with a blended fair value of \$122.61 per share, implying significant upside from its current trading level. This valuation also reveals a wide margin of safety, supported by resilient cash flows, best-in-class gross margins, and a defensively positioned product portfolio. Whether viewed through the lens of dividend reliability or free cash flow generation, Merck offers a compelling opportunity for both growth, and income-oriented investors.

Overall, the valuation results demonstrate that Merck & Co. offers a compelling risk-reward profile across a range of economic and strategic outcomes. The downside appears limited, even under conservative assumptions, while the upside potential (56.41%) remains substantial, particularly if the company continues to deliver innovation through its pipeline and successfully manages the post-Keytruda transition. As a result, the valuation strongly supports a **Buy recommendation**, backed by both intrinsic value and margin of safety.

9. Appendix

Gross margin		ROE		ROA		Asset turnover	
Sales	64168	Net income	17117	Net income	17117	Revenue	64168
COGS	14698	Shareholder equity	46372	Shareholder equity	117106	Assets	117106
GM	77,09%	ROE	36,91%	ROA	14,62%	ATO	54,79%
Current ratio		Debt to Equity		Price to book		Price to earnings	
Current assets	38782	Total debt	34611	Market cap (8/4)	205060	Share price	81
Current liabilities	28420	Total equity	46372	Book value of equity	46372	EPS	6,78
CR	136,46%	DE	74,64%	PB	4,42	PE	11,95

Figure 11: Ratio calculations

Beta	
Variance SPY	0,0001771
Covariance Mrk & SPY	0,0000806
Beta	0,455

Figure 12: Beta calculation

CAPM	
Beta	0,455
Market return	0,1
Risk free rate	0,04339
Risk premium	0,025758
Expected return	6,91%

Figure 13: CAPM calculation

Sales projection	2024	2025	2026	2027	2028	2029	2030	CAGR
human health	27918	27918	29872,26	31963,32	34200,75	36594,8	39156,44	7,00%
Keytruda (50%) of sales	29482	29482	35378,4	40685,16	44753,68	49229,04	54151,95	12,93%
animal health	5877	5877	6347,16	6854,933	7403,327	7995,594	8635,241	8,00%
Global sales	63277	63277	71597,82	79503,41	86357,75	93819,44	101943,6	10,01%

Figure 14: FCFE base case sales projections

FCFE Base case										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	TV
Pharma sales	52005	53583	57400	57400	65250,66	72648,4782	78954,4265	85823,84661	93308,38718	
Animal sales	5625	5625	5877	5877	6347	6855	7403	7996	8635	
Total revenue	57630	59208	63277	63277	71598	79503	86358	93819	101944	
COGS	17411	16126	15193	11073	12530	13913	15113	16418	17840	
Gross profit	40219	43082	48084	52204	59068	65590	71245	77401	84103	
Research & Development	13548	30531	17938	21814	24683	27408	29771	32344	35144	
Selling & general admin	10042	18342	10816	13815	15631	17357	18854	20483	22257	
Restructuring costs	337	599	309	440	498	552	600	652	708	
Depreciation & Amortization	3909	3872	4034	4155	4701	5220	5670	6160	6694	
EBIT	12383	-10262	14987	11980	13555	15052	16350	17762	19301	
Less tax (16.5% tax rate)	2043	0	2473	1977	2237	2484	2698	2931	3185	
EBIAT	10339,805	-10262	12514	10003	11319	12568	13652	14832	16116	
Plus Depreciation & Amortization	3909	3872	4034	4155	4701	5220	5670	6160	6694	
Less CAPEX	4388	3863	3372	3372	8646	9159	5604	6088	6615	
Less changes in NWC	2700	2774	2965	2965	3355	3725	4046	4396	4776	
FCFE (\$USD millions)	7161	-13027	10211	7821	4019	4904	9672	10508	11418	392366
FCFE (\$ USD)	7160626253	-13027113886	10211383407	7821248810	4019041326	4904476957	9672171545	10507889323	11417808085	3,9E+11
PV				7522266974	3759275396	4290975543	7915329563	8043446391	8175062681	2,8E+11
Net PV of FCFE	3,2064E+11									
Stock value	126,93									
EPS (EBIT)	4,90	-4,06	5,93	4,74	5,37	5,96	6,47	7,03	7,64	

Figure 15: FCFE base case valuation

Sales projection								
	2024	2025	2026	2027	2028	2029	2030	Avg growth rate p/a
human health	27918	27918	30430,62	33169,38	36154,62	39408,54	42955,3	9,00%
Keytruda (50%) of sales	29482	29482	35378,4	42454,08	50944,9	58586,63	67374,62	17,97%
animal health	5877	5877	6464,7	7111,17	7822,287	8604,516	9464,967	10,00%
Global sales	63277	63277	72273,72	82734,63	94921,8	106599,7	119794,9	13,62%

Figure 16: FCFE bull case sales projections

FCFE Base case										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	TV
Pharma sales	52005	53583	57400	57400	65809,02	75623,4558	84976,81162	95554,05619	107522,6525	
Animal sales	5625	5625	5877	5877	6465	7111	7822	8605	9465	
Total revenue	57630	59208	63277	63277	72274	82735	92799	104159	116988	
COGS	17411	16126	15193	11073	12648	14479	16240	18228	20473	
Gross profit	40219	43082	48084	52204	59626	68256	76559	85931	96515	
Research & Development	13548	30531	17938	21814	24916	28522	31992	35908	40331	
Selling & general admin	10042	18342	10816	13815	15779	18063	20260	22740	25541	
Restructuring costs	337	599	309	440	502	575	645	724	813	
Depreciation & Amortization	3909	3872	4034	4155	4745	5432	6093	6839	7681	
EBIT	12383	-10262	14987	11980	13683	15664	17569	19720	22149	
Less tax (16.5% tax rate)	2043	0	2473	1977	2258	2585	2899	3254	3655	
EBIAT	10339,805	-10262	12514	10003	11426	13079	14670	16466	18494	
Plus Depreciation & Amortization	3909	3872	4034	4155	4745	5432	6093	6839	7681	
Less CAPEX	4388	3863	3372	3372	8690	9369	6022	6759	7592	
Less changes in NWC	2700	2774	2965	2965	3386	3876	4348	4880	5481	
FCFE (\$USD millions)	7161	-13027	10211	7821	4095	5266	10394	11666	13103	450268
FCFE (\$ USD)	7160626253	-13027113886	10211383407	7821248810	4094742933	5266376872	10393609845	11665884413	13102753093	4,5E+11
PV				7522266974	3830084120	4607605370	8505726650	8929853846	9381470333	3,2E+11
Net PV of FCFE	3,65165E+11									
Stock value	144,56									
EPS (EBIAT)	4,90	-4,06	5,93	4,74	5,42	6,20	6,96	7,81	8,77	

Figure 17: FCFE bull case valuation

Sales projection								
	2024	2025	2026	2027	2028	2029	2030	CAGR
human health	27918	27918	29313,9	30779,595	32318,575	33934,503	35631,229	5,00%
Keytruda (50%) of sales	29482	29482	35378,4	38916,24	42807,864	44948,257	47195,67	9,87%
animal health	5877	5877	6229,62	6603,3972	6999,601	7419,5771	7864,7517	6,00%
Global sales	63277	63277	70921,92	76299,232	82126,04	86302,338	90691,65	7,46%

Figure 18: FCFE bear case sales projections

FCFE Base case										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	TV
Pharma sales	52005	53583	57400	57400	64692,3	69695,835	75126,43875	78882,76069	82826,89872	
Animal sales	5625	5625	5877	5877	6230	6603	7000	7420	7865	
Total revenue	57630	59208	63277	63277	70922	76299	82126	86302	90692	
COGS	17411	16126	15193	11073	12411	13352	14372	15103	15871	
Gross profit	40219	43082	48084	52204	58511	62947	67754	71199	74821	
Research & Development	13548	30531	17938	21814	24450	26304	28312	29752	31265	
Selling & general admin	10042	18342	10816	13815	15484	16658	17930	18842	19800	
Restructuring costs	337	599	309	440	493	530	571	600	630	
Depreciation & Amortization	3909	3872	4034	4155	4657	5010	5392	5667	5955	
EBIT	12383	-10262	14987	11980	13427	14445	15549	16339	17170	
Less tax (16.5% tax rate)	2043	0	2473	1977	2216	2383	2566	2696	2833	
EBIAT	10339,805	-10262	12514	10003	11212	12062	12983	13643	14337	
Plus Depreciation & Amortization	3909	3872	4034	4155	4657	5010	5392	5667	5955	
Less CAPEX	4388	3863	3372	3372	8602	8951	5329	5600	5885	
Less changes in NWC	2700	2774	2965	2965	3323	3575	3848	4044	4249	
FCFE (\$USD millions)	7161	-13027	10211	7821	3943	4546	9198	9666	10158	349059
FCFE (\$ USD)	7160626253	-13027113886	10211383407	7821248810	3943339719	4545605105	9198214511	9665964874	10157573132	3,5E+11
PV				7522266974	3688466672	3976995001	7527461534	7398980698	7272744157	2,5E+11
Net PV of FCFE	2,8731E+11									
Stock value	113,74									
EPS (EBIAT)	4,90	-4,06	5,93	4,74	5,32	5,72	6,16	6,47	6,80	

Figure 19: FCFE bear case valuation

DDM													
Year	2019	2020	2021	2022	2023	2024	2025	2026	2027	2028	2029	2030	TV
dividend/share	2,26	2,48	2,64	2,76	2,96	3,12	3,19	3,25	3,32	3,39	3,45	3,52	3,59
PV							3,06	3,04	2,90	2,77	2,64	2,52	123,28
Sum discounted dividends	16,95												
Discounted terminal value	88,2655399												
Share valuation	105,22												

Figure 20: DDM valuation

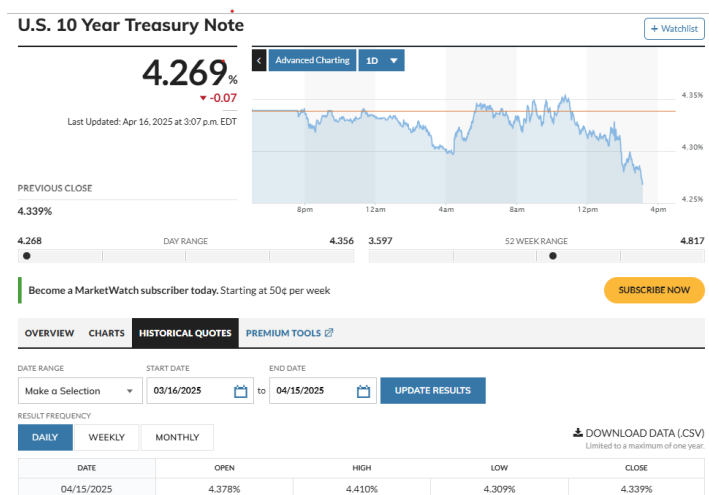


Figure 21: Risk free rate at 15/4/2025

Market Risk (MR)

- **MR1: As the Company's products lose market exclusivity, the Company generally experiences a significant and rapid loss of sales from those products.**
- MR2: The Company faces continued pricing pressure with respect to its products.
- **MR3: The Company faces intense competition from both lower-cost generic products and competitors' products.**
- MR4: The Company may not be able to realize the expected benefits of its investments in emerging markets.
- MR5: Developments following regulatory approval may adversely affect sales of the Company's products.

Operational Risk (OR)

- OR1: The Company's research and development efforts may not succeed in developing commercially successful products and the Company may not be able to acquire commercially successful products in other ways.
- OR2: Failure to attract and retain highly qualified personnel could affect the Company's ability to successfully develop and commercialize products.
- OR3: The Company may experience difficulties and delays in manufacturing certain of its products, including vaccines.
- OR4: Reliance on third-party relationships and outsourcing arrangements could materially adversely affect the Company's business.
- OR5: Negative events in the animal health industry could have a material adverse effect on future results of operations and financial condition of the Company or its Animal Health business.
- **OR6: Biologics and vaccines carry unique risks and uncertainties, which could have a material adverse effect on the Company's future results of operations and financial condition.**
- **OR7: The Company's products, including products in development, cannot be marketed unless the Company obtains and maintains regulatory approval.**

Financial Risk (FR)

- **FR1: Key products generate a significant amount of the Company's profits and cash flows, and any events that adversely affect the markets for its leading products could have a material adverse effect on the Company's results of operations and financial condition.**
- FR2: Unfavorable or uncertain economic conditions, together with cost-reduction measures being taken by certain governments, could negatively affect the Company's operating results.
- FR3: The Company has significant global operations, which expose it to additional risks, and any adverse event could have a material adverse effect on the Company's results of operations and financial condition.
- FR4: The Company is exposed to market risk from fluctuations in currency exchange rates and interest rates.

Legal and Regulatory Risk (LR)

- LR1: The Company is dependent on its patent rights, and if its patent rights are invalidated or circumvented, its business could be materially adversely affected.
- LR2: The Company's success is dependent on the successful development and marketing of new products, which are subject to substantial risks.
- LR3: Climate change or legal, regulatory, or market measures to address climate change may negatively affect the Company's business, results of operations, cash flows, and prospects.
- LR4: Pharmaceutical products can develop unexpected safety or efficacy concerns.
- **LR5: The health care industry in the U.S. has been, and will continue to be, subject to increasing regulation and political action.**
- LR6: The Company is subject to a variety of U.S. and international laws and regulations.
- LR7: The Company is subject to evolving and complex tax laws, which may result in additional liabilities that may affect results of operations and financial condition.
- LR8: Adverse outcomes in current or future legal matters could negatively affect Merck's business.
- LR9: Product liability insurance for products may be limited, cost prohibitive, or unavailable.

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